

Analyzing the Effect of Pitch Arsenal on Strikeout Rate

By Evan Cerne

Introduction

Pitchers in Major League Baseball have reached the highest level of baseball in the world. What gets them there is their ability to get outs against talented hitters. To do this, they rely on an intentionally developed arsenal of pitches. The website Baseball Savant (<https://baseballsavant.mlb.com/>) tracks 11 different pitch types, creating many possible arsenals. While two different pitchers may have the same arsenal, the way that each pitcher deploys their pitches surely differs. Some pitchers focus on creating weak contact, while others aim to strike batters out. It becomes interesting to look at what types of arsenals are most effective at accomplishing these goals. Specifically, this paper explores what is the most effective arsenal for striking out batters.

Previous research has been done modeling the strikeout rate of pitchers. A paper on how the predictability of a pitch sequence influences strikeout rate was written in 2017. In that paper, a model that predicts strikeout rate was created. The predictors in this model were fastball movement, fastball velocity, pitch arsenal and pitch sequence. The model was adjusted based on the handedness of both the batter and the pitcher. It was concluded that fastballs with more vertical movement and higher velocity lead to higher strikeout rates, while greater sequencing predictability leads to lower strikeout rates (Healey and Zhao, 2017).

A 2015 paper (written by one of the coauthors of the previously mentioned paper) modeled the probability of a strikeout occurring in any given plate appearance. An initial model was created using the league strikeout rate, the batter's strikeout rate and the pitcher's strikeout rate as predictors. The response variable was the probability of a strikeout occurring in the plate appearance. The natural log of every variable was taken to achieve this model. This model was shown to be accurate, with improved accuracy achievable through small changes in coefficients depending on the handedness of the batter and pitcher. An interaction term between the pitcher's ground ball rate and the batter's ground ball rate was then added to the model and shown to be significant. There is an intuitive justification for this. Pitchers that induce a high rate of groundballs tend to make hitters swing over their pitches. Batters with high groundball rates tend to swing over the ball. When a pitcher with a high groundball rate faces a batter with a high groundball rate, these tendencies compound to create a higher probability of a swing and a miss. When a pitcher with a high groundball rate faces a hitter that tends to swing under the ball and hit fly balls,

the tendencies cancel out. This leads to a lower probability of a swing and a miss (Healey, 2015).

The analysis in the present paper reveals that fastball type has a main effect on a pitcher's strikeout rate, while secondary type has no effect on a pitcher's strikeout rate. There is also no significant interaction between fastball type and secondary type. The next section is the Methods sections, followed by the Analysis of the data, Conclusion and References.

Methods

Here is the data set, which was compiled using data from Baseball Savant:

https://docs.google.com/spreadsheets/d/1_NX0Pp20-7LWZu5afSNRvd0Zb8r2olyH/export?format=xlsx

There are two factors and one response variable in the dataset. Each factor has two levels. The first factor is the type of fastball a pitcher throws most commonly. The first level is that they throw a four-seam fastball as their most common fastball. The second level is that they throw a sinker or a cutter as their most common fastball. The second factor is the type of secondary offering a pitcher most commonly throws. For the purposes of this paper, a secondary offering is defined to be a pitch that is not a fastball. The first level of this factor is that the pitcher throws a breaking ball as their most common secondary pitch. The second level of this factor is that the pitcher throws an offspeed pitch as their most common secondary offering. An offspeed pitch is defined as a secondary offering that is not a breaking ball. Two factors with two levels each results in four treatment groups. The response variable is the strikeout rate of the pitcher (the percentage of batters that the pitcher strikes out).

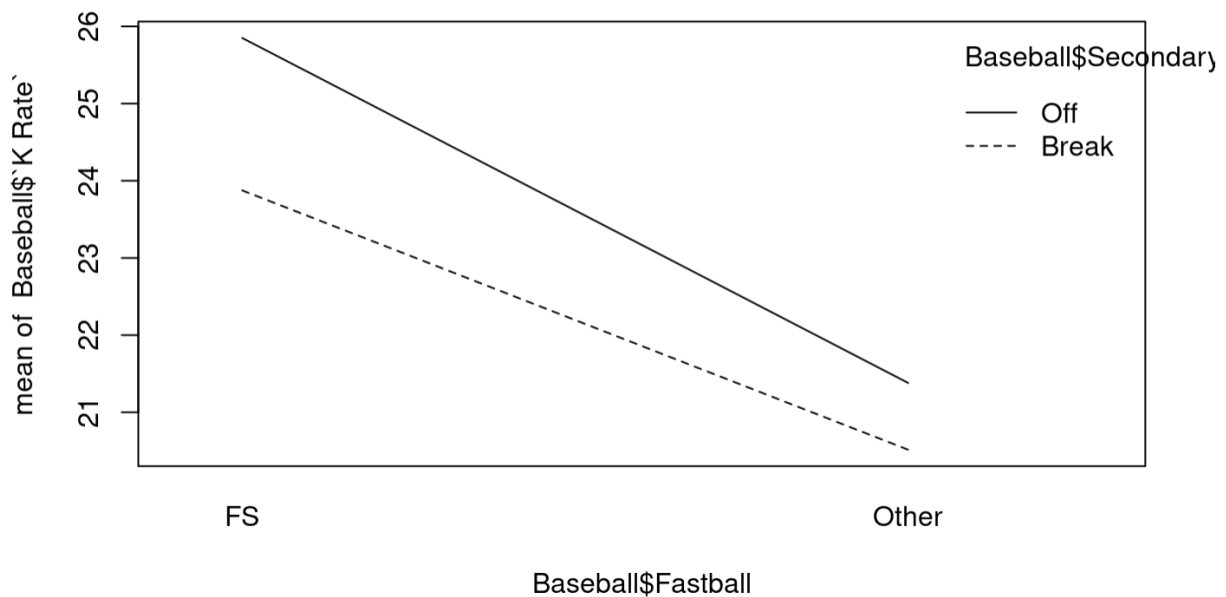
Pitchers that faced at least 250 batters in the 2024 MLB Season were eligible for inclusion in the dataset. There were 277 of these pitchers. They were then organized into the four treatment groups. To ensure a balanced design, 20 pitchers from each treatment group were randomly selected for inclusion in the dataset. In total, 80 pitchers were included in the dataset.

Analysis

An interactive model was created, with the two factors and their interaction as the predictor variables. An Analysis of Variance (ANOVA) was performed on this model. It is shown below:

Factor	F Value	P Value
Fastball	13.2889	0.0004862
Secondary	1.7482	0.1900630
Fastball : Secondary	0.2671	0.6068102

There is a strong Fastball effect ($p=0.0005$), while there is no Secondary effect ($p=0.1901$). There is also no significant interaction ($p=0.6068$). This can be seen in the interaction plot below, as the lines are nearly parallel:

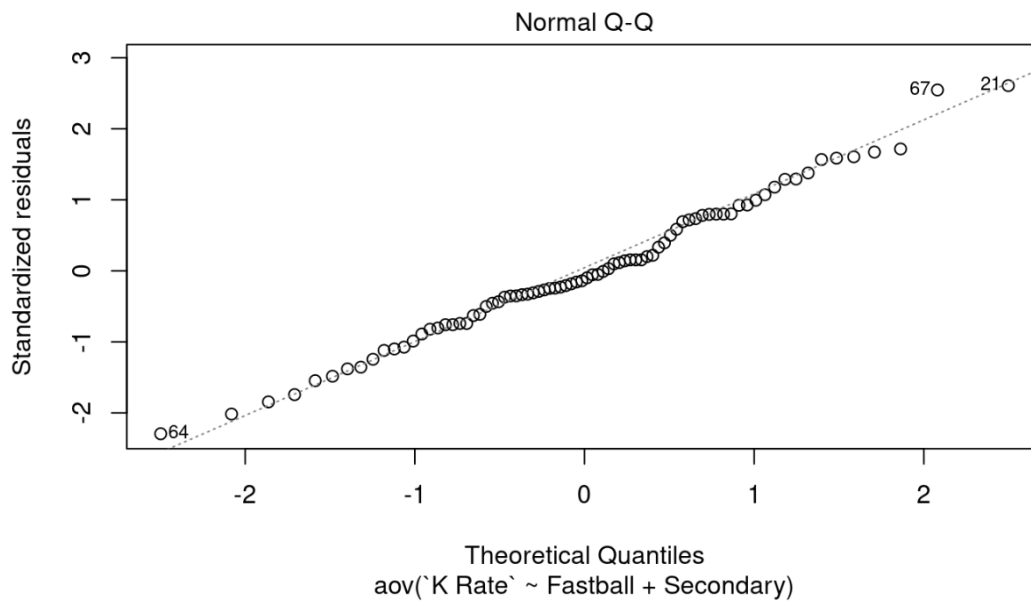


Since there is no interaction, the additive model can be fitted. Here is the ANOVA table for the additive model:

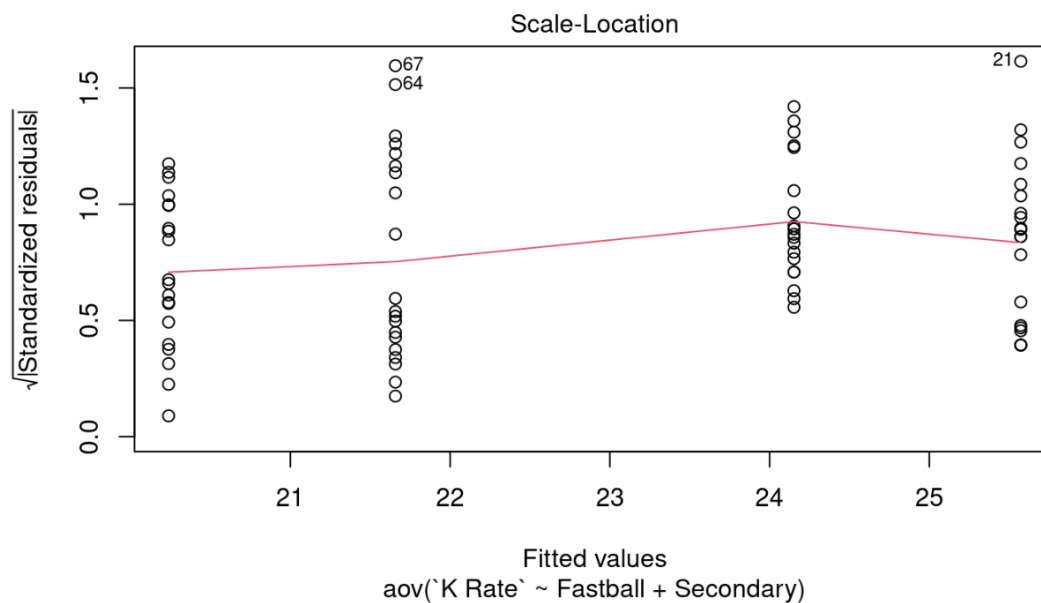
Factor	F Value	P Value
Fastball	13.417	0.0004556
Secondary	1.765	0.1879203

Since there was no interaction between the two factors, the p values for each factor are nearly identical in the additive model. There is still a Fastball effect, but no Secondary effect.

It is important to test the assumptions of Two-Way ANOVA. In the Normal Probability Plot, all points fall closely to the line, indicating normality. The plot is shown below:



The scale-location plot reveals whether the variance of the error is constant. In the plot below, the kernel smoother is relatively flat, indicating that the error does not change depending on the treatment group. This means that the homoskedasticity assumption holds.



The independence assumption is met because each pitcher is present in exactly one of the four treatment groups. Since the assumptions are all met and there is no interaction term, the additive model is the appropriate model to fit for the data set.

For local inference, a confidence interval is created for the Fastball factor, and Bonferroni's adjustment is applied. No confidence interval for the Secondary Factor is computed because there is no Secondary effect. Since there is a pairwise comparison for each factor (2 total pairwise comparisons), a 97.5% confidence interval is used.

The Fastball effect can be interpreted as follows: We can be 97.5% confident that the true difference in K Rate between pitchers who throw sinkers or cutters as their primary fastball and pitchers who throw four-seam fastballs as their primary fastball is between -6.358% and -1.472%. This indicates that throwing a four-seam fastball as the primary fastball in an arsenal is better for striking hitters out.

Conclusion

The analysis of the data reveals that the effect of the type of fastball that a pitcher throws on strikeout rate does not depend on the type of secondary offering that a pitcher throws. The effect of the type of secondary offering that a pitcher throws on strikeout rate does not depend on the type of fastball that a pitcher throws. In fact, the type of secondary pitch a pitcher throws does not have a significant effect on their strikeout rate. However, the type of fastball that a pitcher throws does affect their strikeout rate: Four-seam fastballs are more effective than sinkers/cutters at striking batters out. This result is consistent with a finding from Healey and Zhao (2017). A pitcher's four-seam fastball is typically their fastest pitch, and that paper concluded that fastballs with higher velocity result in higher strikeout rates. However, another finding from Healey and Zhao (2017) contradicts the results from this paper. Sinkers and cutters generally have more vertical movement than four-seam fastballs, and the 2017 paper found that fastballs with more vertical movement lead to a higher strikeout rate (Healey and Zhao, 2017).

It is surprising that this analysis did not reveal a Secondary effect, since breaking balls and offspeed pitches are so fundamentally different. A larger sample size would likely result in a Secondary effect.

References

G. Healey, "Modeling the Probability of a Strikeout for a Batter/Pitcher Matchup," in IEEE Transactions on Knowledge and Data Engineering, vol. 27, no. 9, pp. 2415-2423, 1 Sept. 2015, doi: 10.1109/TKDE.2015.2416735.

Healey, G., & Zhao, S. (2017). Using PITCHf/x to model the dependence of strikeout rate on the predictability of pitch sequences. Journal of Sports Analytics, 3(2), 93-101.
<https://doi.org/10.3233/JSA-170103> (Original work published 2017)